

These practices outline good VR audio design. The audio and source should move in the same direction. If users wear headphones, source should also move to follow head movements. Sound should get louder as the users lean towards their source, even if their avatars are stationary.

Interactive designs

Scale of a virtual environment can be made more natural through atmospheric settings. To look at a far object, look through more air and particles. Make farther objects look less sharp and blurrier than close objects. This gradual fading of the landscape is a clear cue for depth and distance.

Reticles show the centre of focus to orient in space. These are used for movement and interaction with objects. These react accordingly when the users turn their attention to an interactive object. Reticles switch between light and dark modes to adapt their colour(s) to the brightness level of the background.

Post-processing steps in the software development kit (SDK) corrects the distortion of the rendered image caused by lenses in VR headsets.

Software tools and frameworks

It is crucial to get your ideas on paper before getting into any software. It is cheaper, faster and helps to express ideas in lesser time than in software. If you are building a 3D game, you can use Unreal Engine and Cinema 4D for complex animations and renderings. There are three frameworks to choose for VR design: Unreal SDK for high-end headsets, Daydream VR for mid-range VR, and cell phones and Mozilla A-Frame for Web VR.

VR software tracks users head and eye movements constantly. This provides visual feedback, allowing images to move and change with every new perspective. While designing a VR experience, you must anticipate and direct users' every movement.

While developing a VR app, there are two choices for origin: floor and camera. Floor as origin causes view-

point at the same height, off the ground, as it is in real life. For this option, build a solution to scale height according to different people.

Whereas, camera as origin controls height within the virtual world, but by using prediction of the physical floor, which becomes difficult.

Sridhar Sunkand, managing director, EON Reality Pvt Ltd, says, "A VR platform includes creation, virtual training and post-production assistance. VR can be used as a teaching tool to learn faster, remember longer and decide better. We provide multi-user, cross-platform, live and on-demand VR training. We develop, run, manage, access, store, host and distribute augmented reality (AR) and VR applications for education. To prevent simulator sickness, designers should be given knowledge to perform complex tasks. They should show the shortest and safest path between two points to eliminate the time lost to navigate complex environments."

Designing a comfortable experience

During a VR experience, viewing a moving environment, where the body is stationary, confuses the brain. To help ease nausea, provide a fixed reference line that stays as the users move. Such movements must be controlled by the users, or they will become sick.

Accommodative demand (AD) and vergence demand (VD) should be taken care of, for eye comfort while focusing on VR objects. AD adjusts the shape of eye lenses to bring depth plane into focus. VD refers to the rotation of eye to intersect the line-of-sight at a particular depth plane. Place the gameplay elements of VR in the users' immediate line-of-sight.

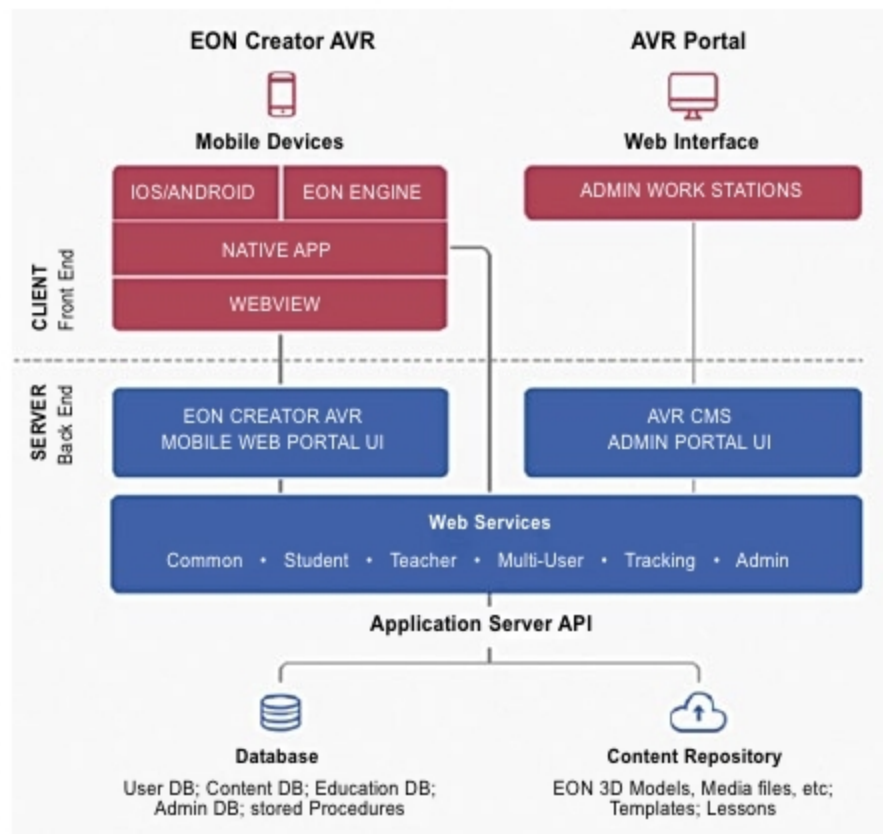
Minimum comfortable viewing distance for VR UI is 0.5 metres. Beyond 10 metres, the sense of 3D stereoscopic depth perception diminishes. Thus, place important content between 0.5 and 10 metres. For gear VR, larger than 42 points ensures a comfortable

reading experience. Font size larger than 25 points gives minimal legibility at fixed z-depth of 4.5 metres.

Latency is the total time between the movement of the user's head and the updated image being displayed on the screen. It includes the times for sensor response, fusion, rendering, image transmission and display response. Experts recommend minimising latency to reduce discomfort, because lag between head movements and corresponding updates on the display can lead to sensory conflicts and errors in the vestibular-ocular reflex.

Threshold for compelling VR must be at or below 20 milliseconds of latency. When latency exceeds 60 milliseconds, disjunction between one's head motions and the motions of the virtual world start to feel out of sync, causing discomfort and disorientation.

Ankur Gupta, head - marketing, Ken Research Pvt Ltd, says, "As per the research done by our company, technology advancement, correct positioning, tapping into new geography and focusing on mobile applications will aid the global VR market. Global VR market is expected to register a positive CAGR of around 54.6 per cent during 2017-2022. Markets will have a remarkable positive impact with expected increase in the number of VR arcades worldwide." **EFY**



EON VR design platform architecture (Credit: Sridhar Sunkand, EON Reality)